

Article

Direct C-C Coupling of CO and the Methyl Group from CH Activation through Facile Insertion of CO into Zn-CH σ -bond

Yuntao Zhao, Chaonan Cui, Jinyu Han, Hua Wang, Xinli Zhu, and Qingfeng Ge

J. Am. Chem. Soc., **Just Accepted Manuscript** • DOI: 10.1021/jacs.6b04446 • Publication Date (Web): 23 Jul 2016

Downloaded from <http://pubs.acs.org> on July 23, 2016

Just Accepted

"Just Accepted" manuscripts have been peer-reviewed and accepted for publication. They are posted online prior to technical editing, formatting for publication and author proofing. The American Chemical Society provides "Just Accepted" as a free service to the research community to expedite the dissemination of scientific material as soon as possible after acceptance. "Just Accepted" manuscripts appear in full in PDF format accompanied by an HTML abstract. "Just Accepted" manuscripts have been fully peer reviewed, but should not be considered the official version of record. They are accessible to all readers and citable by the Digital Object Identifier (DOI®). "Just Accepted" is an optional service offered to authors. Therefore, the "Just Accepted" Web site may not include all articles that will be published in the journal. After a manuscript is technically edited and formatted, it will be removed from the "Just Accepted" Web site and published as an ASAP article. Note that technical editing may introduce minor changes to the manuscript text and/or graphics which could affect content, and all legal disclaimers and ethical guidelines that apply to the journal pertain. ACS cannot be held responsible for errors or consequences arising from the use of information contained in these "Just Accepted" manuscripts.



ACS Publications

**Direct C-C Coupling of CO₂ and the Methyl Group from CH₄ Activation through Facile
Insertion of CO₂ into Zn-CH₃ σ-bond**

Yuntao Zhao,[†] Chaonan Cui,^{†,‡} Jinyu Han,[†] Hua Wang,[†] Xinli Zhu^{,†} and Qingfeng Ge^{*,†,‡}*

[†]Collaborative Innovation Center of Chemical Science and Engineering, School of Chemical
Engineering and Technology, Tianjin University, Tianjin 300072, China

[‡]Department of Chemistry and Biochemistry, Southern Illinois University, Carbondale, Illinois
62901, United States

Corresponding Authors

*Tel.:86-22-27890859 Email: xinlizhu@tju.edu.cn (X.Z.)

*Tel.:1-618-453-6406 Email: qge@chem.siu.edu (Q.G.)

ABSTRACT

Conversion of CO₂ and CH₄ to value-added products will contribute to alleviating the green-house gas effect but is a challenge both scientifically and practically. Stabilization of the methyl group through CH₄ activation and facile CO₂ insertion ensure the realization of C-C coupling. In the present study, we demonstrate the ready C-C coupling reaction on a Zn-doped ceria catalyst. The detailed mechanism of this direct C-C coupling reaction was examined based on the results from density functional theory calculations. The results show that the Zn dopant stabilizes the methyl group by forming a Zn-C bond, thus hindering subsequent dehydrogenation of CH₄. CO₂ can be inserted into the Zn-C bond in an activated bent configuration, with the transition state in the form of a three-centered Zn-C-C moiety and an activation barrier of 0.51 eV. The C-C coupling reaction resulted in the acetate species, which could desorb as acetic acid by combining with a surface proton. The formation of acetic acid from CO₂ and CH₄ is a reaction with 100% atom economy and the implementation of the reaction on a heterogeneous catalyst is of great importance to the utilization of the greenhouse gases. We tested other possible dopants including Al, Ga, Cd, In and Ni and found a positive correlation between the activation barrier of C-C coupling and the electronegativity of the dopant although C-H bond activation is likely the dominant reaction on the Ni-doped ceria catalyst.

KEYWORDS: Zn-doped ceria, Zn-CH₃ species, CO₂ and CH₄ conversion, C-C coupling, electronegativity of dopant, DFT

1 INTRODUCTION

2 Increased consumption of fossil fuels since the industrial revolution has resulted in a steady
3 increase of the CO₂ concentration in the atmosphere. CO₂ is the main contributor to the global
4 warming which poses a devastating threat to the sustainability and prosperity of the society.¹
5 Carbon capture and storage (CCS), once implemented, could stabilize the atmospheric CO₂ level,
6 but the high cost hinders its deployment.² Conversion of CO₂ to high-energy-density and other
7 high value products addresses both climate change and fossil fuel shortage. Due to the inertness
8 of CO₂, its conversion requires energy inputs in various forms, including thermal, electrical,
9 photo or more energetic reactants. For example, electricity is used in electrochemical reduction,³
10 solar energy in photo-reduction reaction,⁴ hydrogen in methanol synthesis and methanation
11 reaction⁵ as well as CH₄ in the reforming reaction.⁶ Abundant shale gas reserves make CH₄ a
12 viable source of both energy and hydrogen for CO₂ conversion.

13 Most of the previous studies focused on the traditional syngas route, i.e. using CO₂ and CH₄
14 to produce CO and hydrogen, followed by either the methanol synthesis or Fischer-Tropsch
15 processes to produce high value chemicals. The syngas route has the advantage of easy
16 integration with the existing technology, but the syngas production from CO₂ reforming of CH₄
17 is energy-intensive. There have been attempts at directly synthesizing C₂ hydrocarbons from CH₄
18 and CO₂ using a La₂O₃/γ-Al₂O₃ catalyst, but the results showed that both carbon atoms in the
19 product came from CH₄.⁷

20 Recently, Banerjee and coworkers identified a sustainable C-C bond formation route
21 through the reaction of CO₂ with deprotonated C-H bonds, which is simple and applicable to

1 economical production.⁸ However, facile C-C bond formation from CO₂ and CH₄ is still
2 challenging and potentially beneficial since co-conversion of CO₂ and CH₄ into acetic acid has
3 an atom economy 100%, in which the formation of acetate species (CH₃COO*) through C-C
4 coupling is dominant.⁹ The reaction has been accomplished experimentally under homogeneous
5 conditions, catalyzed by vanadium complexes¹⁰ and Pd(OAc)₂/Cu(OAc)₂/K₂S₂O₈/CF₃COOH,¹¹
6 but product separation at the end of the reaction presented a great challenge. So far, many
7 heterogeneous catalysts, such as Cu/Co-based metal oxide,¹² Pd/C and Pt/Al₂O₃,¹³
8 V₂O₅-PdCl₂/Al₂O₃,¹⁴ Pd/SiO₂ and Rh/SiO₂,¹⁵ have been attempted for the co-conversion of CO₂
9 and CH₄ into acetic acid. The formation of acetic acid under plasma conditions was also
10 investigated.¹⁶ In each case, however, the result shows a broad product distribution, poor
11 selectivity, and a low yield of acetic acid. This could be a result of the failure to stabilize the
12 methyl radical and a high C-C coupling barrier.

13 Interestingly, the methyl radical could be successfully stabilized on metal sites of bulky
14 ligands and carboxylation could be realized through CO₂ inserting into M-CH₃ bond in organic
15 synthesis.¹⁷ For example, Darensbourg and coworkers reported the reaction of CO₂ insertion into
16 *cis*-CH₃W(CO)₅.^{17b} They found that at a low CO₂ pressure, the insertion step occurs over days
17 along with the decomposition of the product. Johansson et al. used approximately 30 equivalents
18 of CO₂ in a coupling reaction with PCP^(tBu)Pd-CH₃ (PCP =
19 2,6-bis[(di-*tert*-butylphosphino)methyl]-phenyl) and tested the product PCP^(tBu)PdOAc using
20 NMR.^{17c} In a recent study, inserting CO₂ to (PCP)Ni-CH₃ was shown to have a very high barrier
21 of 1.32 eV (all energies involved in this article are converted into eV for comparison).^{17f} Efficient

formation of C-C bond is the major challenge for producing compounds using CO₂. Although these CO₂ insertion reactions are slow, their realization in homogeneous synthesis provides inspiration for implementing a similar process under heterogeneous condition.

Recently, selective activation of CH₄ has been achieved on zinc-exchanged MFI or ZSM-5 zeolites under heterogeneous condition.^{9a, 18} The methyl from CH₄ dissociation was stabilized on the zinc site in the form of $-\text{[ZnCH}_3\text{]}^+$ species, observed by IR and NMR. C-C bond formation through CO₂ insertion into the M-C σ -bond is expected to be the key step in a heterogeneously catalyzed C-C coupling reaction. A theoretical study on the Au-ZSM-5 catalyst showed that C-C coupling could be achieved by CO₂ insertion into the Au-CH₃ bond, however, the coupling barrier of 2.88 eV makes the reaction almost impossible in practice.¹⁹ Importantly, the C-C bond could form through insertion of CO₂ into the organozinc reagents catalyzed by Pd(OAc)₂, [Ni(PCy₃)₂]₂(N₂), Ni(acac)₂ or in the presence of MgCl₂ or LiCl in the solvent.²⁰ Also, CO₂ insertion into the Zn-H bond could be readily accomplished.²¹

Herein, we report the use of a Zn-doped ceria as a catalyst to facilitate the coupling of CO₂ and methyl from CH₄ activation through insertion of CO₂ into the Zn-CH₃ bond. The ability of Zn²⁺ in stabilizing CH₃ to hinder the further oxidation^{9a, 18} and the ceria in activation of CO₂ is key to implement the reaction.²² We use the density functional theory (DFT) method to track the elementary steps and elucidate the reaction mechanism. We showed that the C-C coupling reaction is facile and desorption of acetic acid following the formation of the acetate species depends strongly on the reaction environment. We also explored other metals as dopant in a ceria-based catalyst and found a correlation between the barrier of the coupling reaction and the

electronegativity.

METHODS

Spin-polarized calculations were performed at the DFT+ U level using the Vienna ab initio simulation package (VASP).²³ In order to properly localize the Ce 4f states, the Hubbard-like U correction has been used in combination with the Perdew-Burke-Ernzerhof (PBE) functional.²⁴ The effective U parameter is defined as the difference between the Coulomb U and the exchange J terms. A value of 5.0 eV was set for the U parameter, following our previous work,²⁵ and consistent with the values used by others.²⁶ Projector-augmented wave (PAW) potentials were used to describe the effective cores.²⁷ The valence electrons of all atoms were expanded in a plane wave basis set with a cutoff energy of 400 eV. The atomic structures were relaxed using either the conjugate gradient algorithm or the quasi-Newton scheme as implemented in the VASP code until the forces were less than 0.02 eV/Å for all unconstrained atoms.

We chose the ceria (111) surface and used a slab consisting of 48 O atoms and 24 Ce atoms. Those atoms were distributed in three O-Ce-O trilayers and separated by a vacuum space of 15 Å. Zn-doped ceria (111) surface was constructed by substituting one Ce atom with a Zn atom in the supercell, and oxygen vacancies in the surface layer were created by removing two O atoms in the para-position next to the replaced Ce atom. The atoms in the bottom layer were fixed at their corresponding bulk positions, and the top two layers together with the adsorbates were allowed to relax during the calculation. A (2×3×1) k-point grid was used to generate the K-points.

The adsorption energy was defined:

$$E_{\text{ads}} = E_{(\text{adsorbate/slab})} - E_{(\text{adsorbate})} - E_{(\text{bare slab})} \quad (1)$$

where $E_{(\text{adsorbate/slab})}$, $E_{(\text{adsorbate})}$, and $E_{(\text{bare slab})}$ represent the total energies of surface slab with the adsorbate, the isolated adsorbate molecule, and the optimized bare slab, respectively. A positive value corresponds to an endothermic process, whereas a negative value indicates the process is exothermic. Normal mode harmonic analysis was applied to stable and transition states. The transition state was confirmed with only one imaginary mode.

RESULTS AND DISCUSSION

Bulk and Zn-doped Ceria

The optimized lattice parameter of bulk ceria is 5.437 Å, consistent with the experimental values²⁸ and previous computational studies.^{26b, 29} We also calculated the dissociative adsorption energy of methane on pure ceria (111) surface and obtained a value of -0.78 eV, in good agreement with the previous reports.^{26a, 30} The surface slab was modeled using a (2×4) surface supercell. The Zn-doped ceria was constructed by replacing a Ce atom with one Zn atom and removing two oxygen atoms in the para-position next to the replaced Ce atom, as shown in Figure 1.

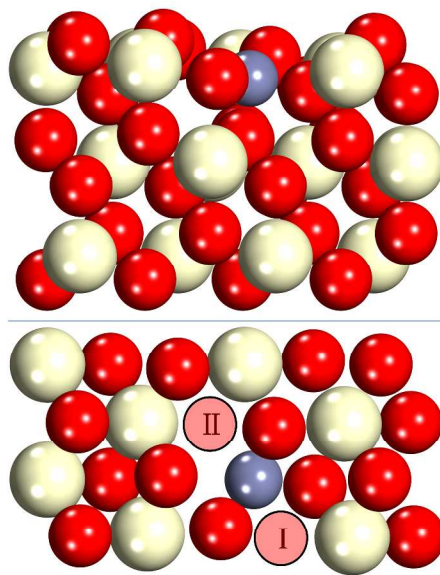


Figure 1. Relaxed structure of the Zn-doped ceria surface, side view (upper) and top view (lower). Surface and subsurface oxygen vacancies are labeled as I and II, respectively. In this and subsequent figures, Ce is ivory, O is red and Zn is purple.

We calculate the energy of oxygen vacancy formation (E_f) according to the following formula:

$$E_f = E_{(\text{Zn}_x\text{Ce}_{1-x}\text{O}_{2-\delta})} + E_{(\text{O}_2)} - E_{(\text{Zn}_x\text{Ce}_{1-x}\text{O}_2)} \quad (2)$$

where $E_{(\text{Zn}_x\text{Ce}_{1-x}\text{O}_{2-\delta})}$, $E_{(\text{O}_2)}$, and $E_{(\text{Zn}_x\text{Ce}_{1-x}\text{O}_2)}$ are the energies of Zn-doped ceria slab with the oxygen vacancies, gas-phase O_2 , and the slab without oxygen vacancies after doping Zn into the lattice, respectively. The calculated vacancy formation energy is 1.21 eV, much lower than that of the pure ceria (111). We point out that the value of 1.21 eV is calculated on the basis of direct desorption of oxygen into the gas phase. The creation of oxygen vacancy is expected to be more facile if assisted by reducing reagents such as H_2 or CO , as shown in the case of In_2O_3 .³¹ The results demonstrated that doping Zn in CeO_2 facilitates oxygen vacancy formation.

Dissociative Adsorption of CH₄

The optimized structure of dissociative adsorption of CH₄ is depicted in Figure 2.

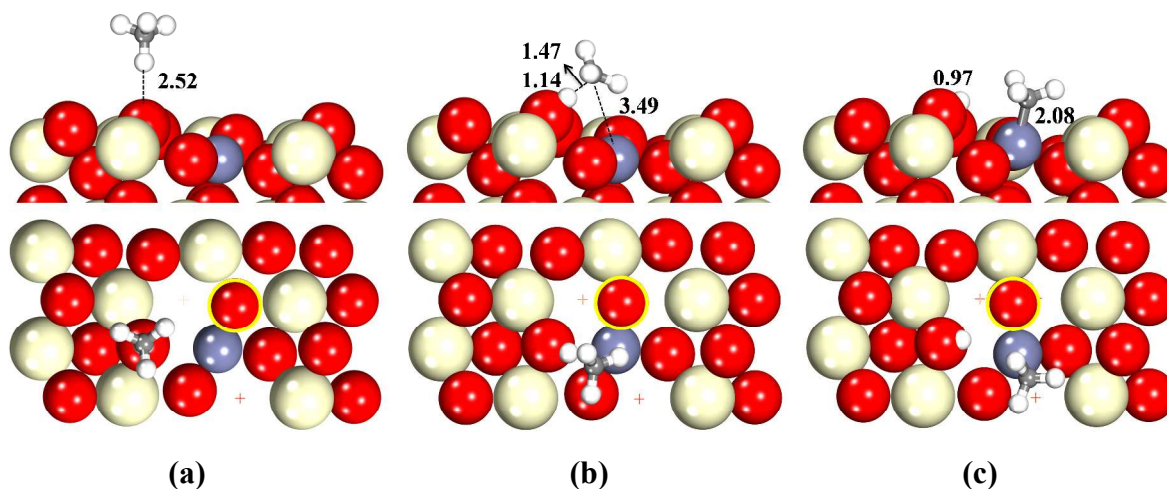


Figure 2. Side view (upper) and top view (lower) of initial (a), transition (b), and final (c) states for dissociative adsorption of CH₄. C and H are marked as gray and white. Distances of H-O, H-C and C-Zn are in Å. O atom with the yellow circle (the same label in the following figures) moves slightly towards the subsurface oxygen vacancy.

Methane is physisorbed on the surface, as shown in Figure 2(a), with the H of CH₄ pointing to the surface O at a distance of 2.52 Å and adsorption energy of -0.04 eV. The physisorbed CH₄ is the initial state for C-H bond activation. In the transition state shown in Figure 2(b), the C-H bond is elongated to 1.47 Å from 1.09 Å of the free CH₄ molecule. After overcoming a small barrier of 0.36 eV, the methyl and H atom became co-adsorbed on the surface. In the final state, the methyl group binds to the zinc site through a Zn-C bond of 2.08 Å. The combination of methyl with a metal ion prevented its further dehydrogenation or oxidation and made it susceptible for the subsequent C-C coupling reaction.^{9a, 32} On the Zn-doped ceria surface, the H atom becomes a proton by forming a O-H bond with a bond length of 0.97 Å. The formation of

the OH species generated a surface Brønsted acid site. The hydroxyl tilts towards the zinc center. At the same time, one of the oxygen atoms (marked with yellow circle in Figure 2) next to the zinc site moved towards the subsurface oxygen vacancy site. The surface methyl species has its Zn-C bond tilted towards the oxygen vacancy. The three C-H bonds of the surface methyl species remain at the same length of 1.10 Å, close to the gas phase value of 1.09 Å. The dissociative CH₄ adsorption energy on the surface is -0.53 eV with respect to the free molecule and bare slab.

To further understand the interaction between methyl species and the surface, we analyzed the projected density of states of Zn and C and plotted the results in Figure 3.

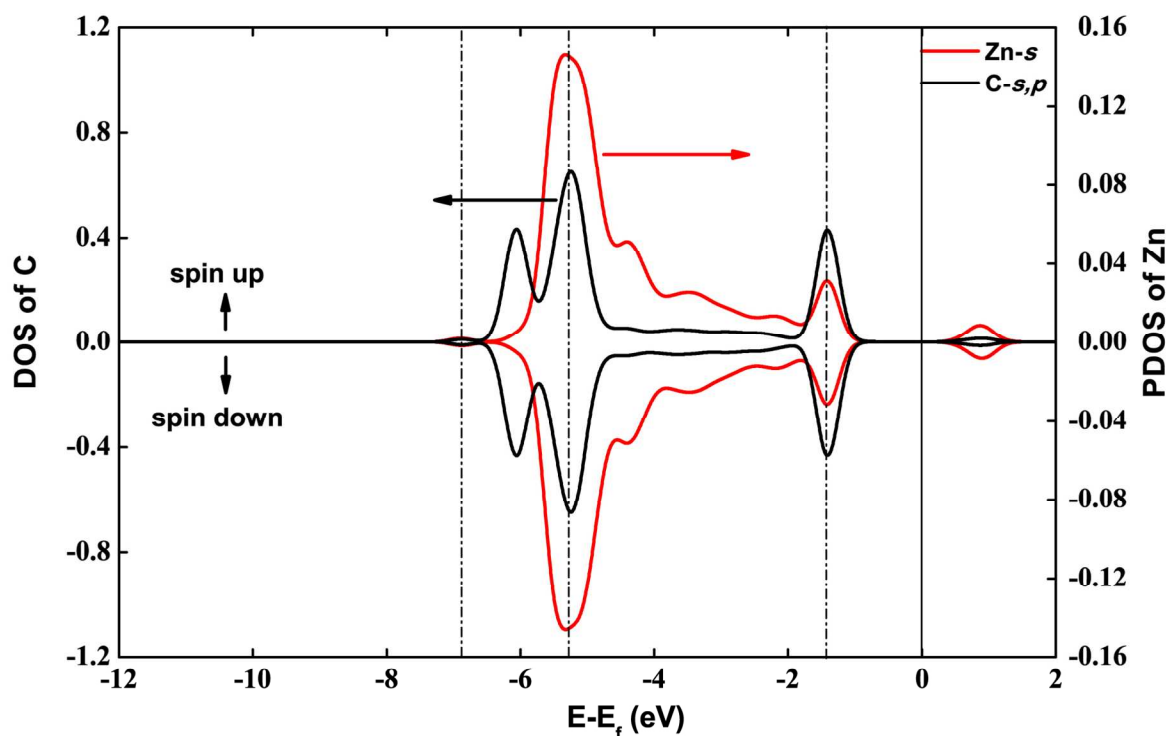


Figure 3. PDOS of Zn (red line) and DOS of C (black line). The Fermi level is indicated by a solid line and the dotted dash lines represent the overlaps between Zn and C.

As shown in Figure 3, significant overlaps between *s*, *p* orbitals of C and *s* state of Zn

1 suggest the formation of a Zn-C bond and stabilize the adsorbed methyl. This observation is in
2 accordance with previous reports that the heterolytic dissociation of the C-H bond in CH₄ could
3 be enhanced by the electron-donation interaction between CH₄-σ(C-H) and Zn 4s.¹⁸

C-C Coupling from -CH₃ and CO₂

6 Starting from the co-adsorbed methyl and H atom on the zinc center and the surface oxygen
7 site, respectively, we mapped out the C-C coupling pathway by introducing CO₂. The highly
8 oxidized carbon of CO₂ makes its insertion into the M-C bonds likely initiated by the
9 nucleophilic attack.³³ Figure 4 shows the transition state of the C atom of the adsorbed methyl
10 with the approaching CO₂.

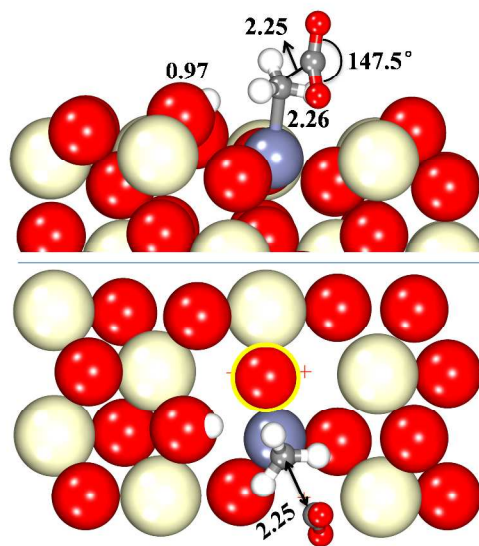


Figure 4. Transition state of C-C coupling. Side and top views are given from top to bottom. Distances are in Å.

In this transition state, the CO₂ molecule is bent from its linear configuration, with a <OCO angle of 147.5°. Meanwhile, both C-O bonds are elongated, to 1.19 Å and 1.24 Å, respectively, from their gas phase value of 1.17 Å. In this process, the O atom (with the yellow circle) stays almost at the same position as that in Figure 2. The Zn-C distance increases by 0.18 Å to 2.26 Å. Consequently, the C atom tends to lose its *sp*³ hybridization as indicated by the planar nature of the three H atoms and C. On the other hand, the hydroxyl group was not affected, remaining in the same position with the same bond length of 0.97 Å. In the transition state, the distance between the two C atoms reaches a value of 2.25 Å and activation of CO₂ occurs via the orbital interaction between CO₂ and the Zn-CH₃ complex as plotted in Figure 5 of the projected density

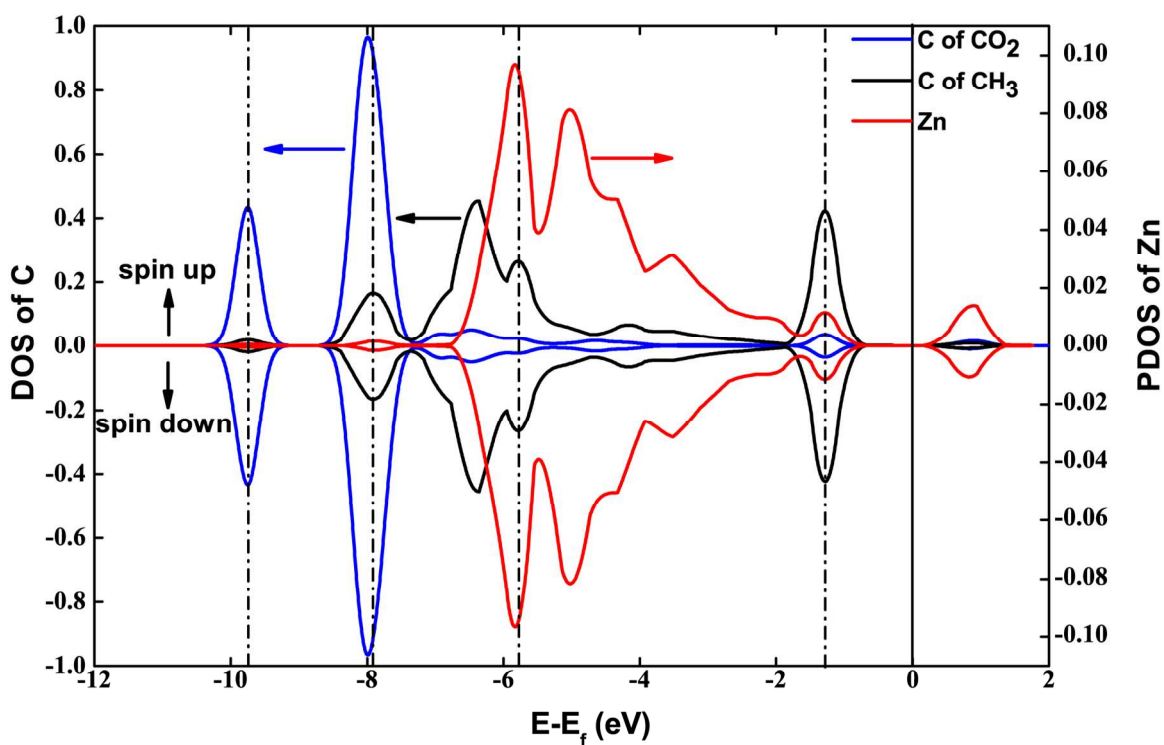


Figure 5. PDOS of Zn (red line); DOS of C of CO₂ (blue line) and C of CH₃ (black line). The solid line represents the Fermi level and the dotted dash lines indicate the overlaps among these three atoms.

of states. As shown in Figure 5, overlaps between the *s* state of Zn and the *s*, *p* states of the two C atoms indicate the formation of the three-center bond in the Zn-C-C moiety.

The formation of the Zn-C-C moiety indicated that the activated CO₂ can insert into the Zn-C bond of zinc-methyl species to form a C-C bond. This heterogeneously catalyzed process has its analogue in organometallics where carboxylic acid or its derivatives could be synthesized through the carboxylation from alkylzinc reagents and CO₂.^{20b-d}

Unlike typical insertion of CO₂ into M-C bond via a 4-centered transition state, in which M-O and C-C bonds are being formed simultaneously with breaking the M-C bond,^{17d, 17e} CO₂ inserts into the Zn-C bond via a S_E2 mechanism through CO₂ attacking the methyl species from the open side of the Zn-CH₃ bond. This observation is consistent with insertions of CO₂ into Pd-CH₃³⁴ and Ni-CH₃.^{17f} Also, in the transition state, the nearly coplanar structure of CH₃ is in line with CO₂ insertion into Ni-CH₃.^{17f}

According to the homogeneous-based catalysis studies, the properties of the ligands, the interaction mechanism in the transition state, the M-CH₃ bond strength and the electrophilicity of CO₂, could all influence the C-C coupling.^{17a-e, 35} The enhancement of nucleophilicity of Zn-CH₃ species could facilitate a facile insertion step. The oxygen vacancies on the catalyst surface increase the electron density at the metal center. For the Zn-CH₃ species, the carbon center of CH₃ has a net charge of -0.34 |e| compared to that in the CH₄ molecule. Moreover, the relaxed Zn-CH₃ bond length (2.08 Å) is longer than that reported in the reference (1.93 Å).^{18a} The high electron density of Zn-CH₃ species and longer Zn-C bond length are consistent with having a methyl group with a longer M-CH₃ distance. A metal center with high electron density is

1 expected to have an enhanced reactivity towards CO₂.^{17a, 17c} In fact, our results show that the
2 activation barrier of C-C coupling is only 0.51 eV on the Zn-doped ceria (111) surface. In this
3 case, the doped Zn atom acted as an active center for the coupling reaction and can be viewed as
4 a single-atom catalyst.³⁶ Despite following the same S_E2 mechanism, the activation barriers for
5 CO₂ insertion into Pd-CH₃ and Ni-CH₃ are 0.83 eV and 1.32 eV, respectively,^{17f, 34} significantly
6 higher than that for insertion into Zn-CH₃.

7 We note that this energy included the zero-point energy correction but not van der Waals
8 (vdW) effect. Ye and coworkers showed that the reaction energy and activation barrier of an
9 elementary step remained almost the same with and without including van der Waals
10 corrections.³⁷ This low activation barrier indicates that coupling the C atom of the Zn-bound
11 methyl species and CO₂ is feasible. Although the Zn-CH₃ species is not surrounded by bulky
12 ligands, which may donate electrons to improve the reactivity of M-C bond,^{17c} the S_E2
13 mechanism resulted in a large <ZnCC angle of 125.4°. This large Zn-C-C angle, together with
14 the sufficient overlaps between orbitals, resulted in a lower activation barrier than the previously
15 reported value for insertion into the Cu^I-CH₃ bond.^{17d} Moreover, the transition metals employed
16 previously, including Pd, Ni, Cu, Ir and Rh, have higher electronegativity than Zn.

17 Direct C-C coupling through insertion of CO₂ into Zn-C σ-bond leads to the formation of an
18 acetate species which adsorbs on the surface in the form of a bidentate pincer through two O
19 atoms interacting with the surface Ce atoms, at O-Ce distances of 2.63 Å and 2.80 Å,
20 respectively. The reaction pathway from CH₄ (g) and CO₂ (g) to acetate species is given in
21 Figure 6.

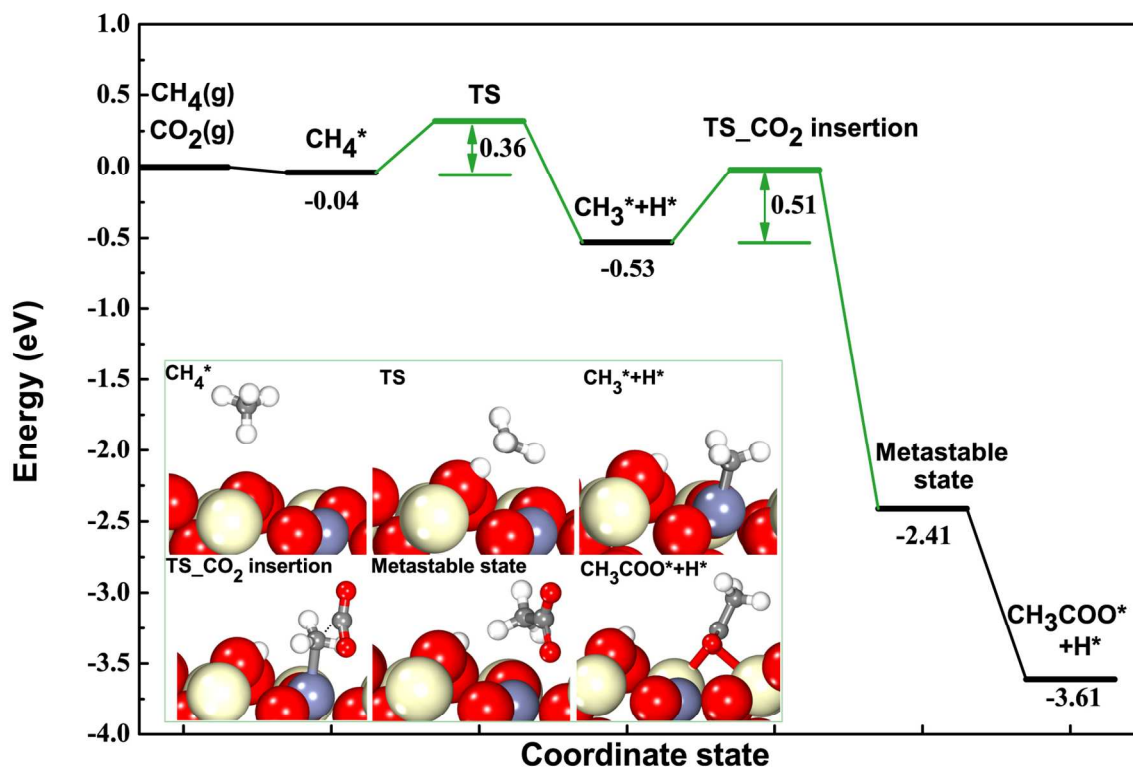


Figure 6. Potential energy profile from gaseous CH₄ and CO₂ to acetate species through C-C coupling. A* represents the adsorption state of species A on the surface.

As shown in Figure 6, methyl is stabilized as zinc-methyl complex after overcoming a barrier of 0.36 eV. The barrier for forming the C-C bond by coupling the methyl species and CO₂ is 0.51 eV, indicating a superior catalytic activity of the active site. Once the acetate species is formed, it transforms to a state where the acetate species lies parallel to the surface. The insertion step is irreversible due to the strong exothermicity of 3.08 eV. The migration of O atom (marked with the yellow circle) contributes -1.23 eV. According to this mechanism, CO₂ comes from gas phase, making the overall reaction of CH₄ and CO₂ follow an Eley-Rideal mechanism.³⁸

Formation of the Product

Production of Aldehyde

Acetate species is reported as a very stable intermediate, up to 500 K before decomposing or desorbing.³⁹ Thus, protonation or deoxygenation of the acetate species is a challenge. We investigated the possible reaction pathways of producing CH_3CHO^* or CH_3COOH^* from co-adsorbed CH_3COO^* and H^* . The relaxed structures are provided in Table S1.

Since the deoxygenation of acetate species to CH_3CO^* is inhibited due to the strong C-O bond, we explored hydrogen-assisted reaction pathway of forming CH_3CHO^* from CH_3COO^* . Our results showed that an energy cost of 2.58 eV is needed to hydrogenate the acetate species to CH_3CHOO^* . The following deoxygenation step to CH_3CHO^* is still 0.73 eV endothermic. On the other hand, if the acetate species eliminates an OH to form CH_2CHO^* , an energy of 2.55 eV is needed. The subsequent hydrogenation of CH_2CHO^* to CH_3CHO^* has a reaction energy of 0.76 eV. These results show that the direct production of aldehyde from the surface acetate species is unlikely due to the strong endothermicity of the reaction steps involved.

Production of Acetic Acid

We mapped out the potential energy profile for production of acetic acid in Figure 7. The black line in Figure 7 follows the two-step formation of acetic acid. The energy cost of transferring the proton from dissociative CH_4 adsorption to the adjacent O site is 0.64 eV. The following combination of the acetate species with the proton to adsorbed acetic acid is 1.30 eV

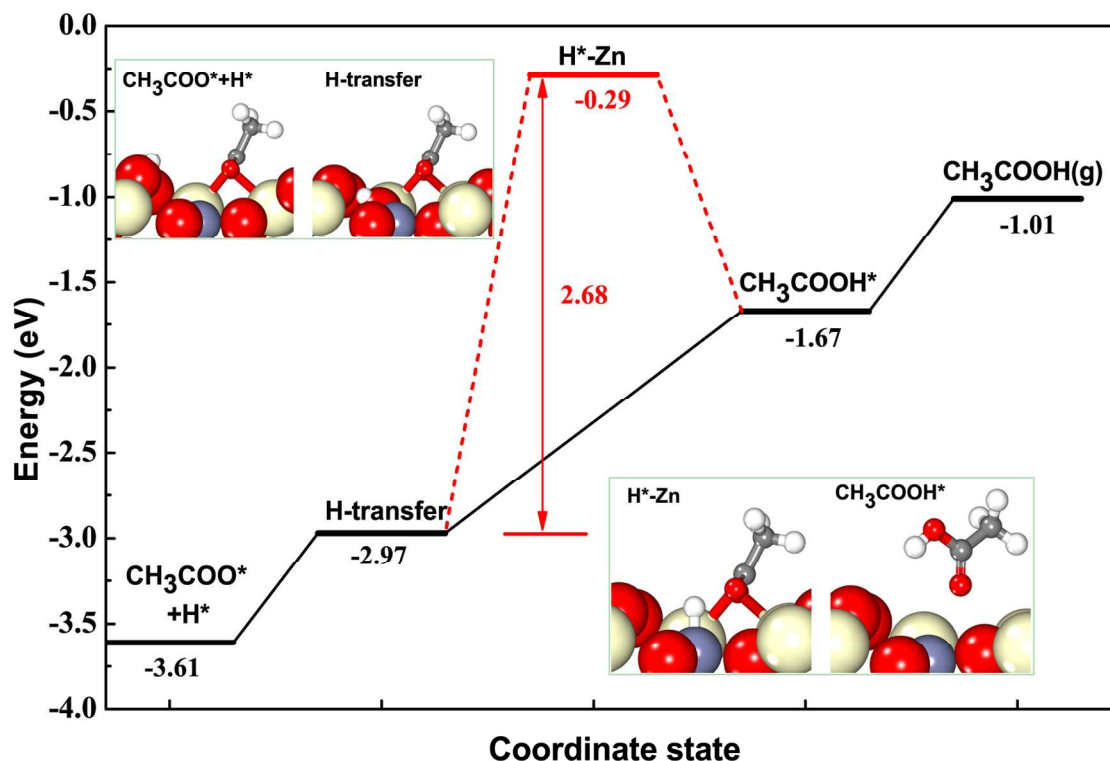


Figure 7. Potential energy profile from acetate species to acetic acid. A* represents the adsorption state of species A on the surface.

1 endothermic. We also considered the possibility of transferring the proton to the zinc site before
 2 adding to the acetate species to form the adsorbed acetic acid, as depicted by the red line in the
 3 figure. We note although the reaction of the acetate species with the H^* on zinc site is strongly
 4 exothermic, the high endothermic step of transferring proton to the zinc site makes this reaction
 5 pathway impossible. Therefore, the production of acetic acid from the acetate species is likely to
 6 follow the pathway shown in black line in Figure 7. The desorption energy of CH_3COOH^* is
 7 0.66 eV. We can see from these results that the steps involved in acetic acid formation from the
 8 acetate species are all endothermic. These steps will likely impede the turnover from the
 9 reactants to the product. In practical implementation, this reaction may be coupled with the
 10 oxidation of CH_4 , which would provide the needed energy for the production of acetic acid from

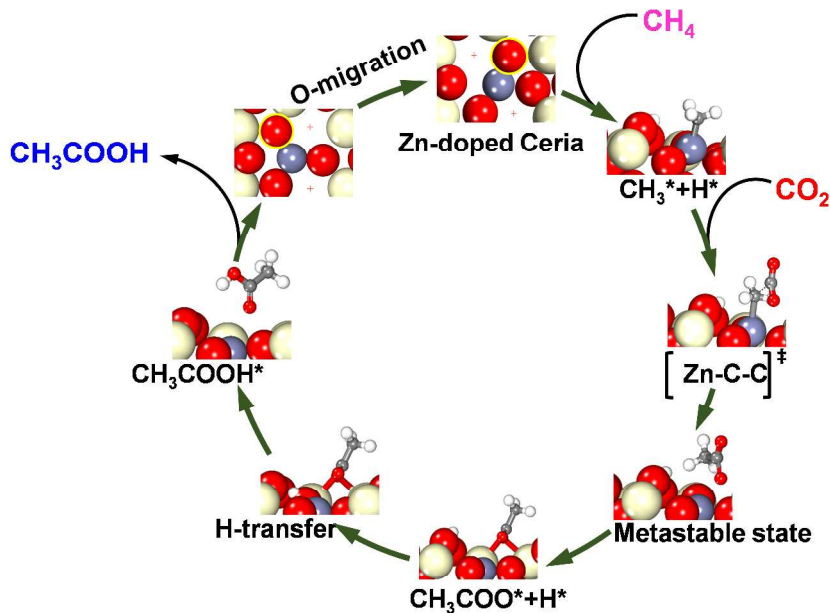
CO₂ and CH₄.

We point out that the catalyst we used has an energy variation of 1.23 eV during the reaction process, attributed to oxygen vacancy activation. Before this step was taking into account, the overall reaction energy from CH₄ and CO₂ to acetic acid is -1.01 eV. When this energy is included in the calculation, the overall reaction energy from gas phase CH₄ and CO₂ to a gaseous acetic acid is 0.22 eV, consistent with the standard thermodynamic data.

Catalytic cycle of C-C coupling

The complete catalytic cycle of Zn-doped ceria catalyzed C-C coupling is shown in Scheme

1.



Scheme 1. Catalytic cycle of production of CH₃COOH from CH₄ and CO₂ through Zn-doped ceria catalyzed C-C coupling. A* represents the adsorption state of species A on the surface.

As shown in Scheme 1, methyl is stabilized as zinc-methyl complex following dissociative adsorption of CH₄. Gaseous CO₂ attacks the Zn-CH₃ species from the open side, resulting into a three-centered Zn-C-C moiety and making the insertion of CO₂ into Zn-C σ -bond a S_E2 mechanism. The barrier for the coupling step is 0.51 eV. The resulting acetate species from the coupling reaction is very stable and its desorption in the form of acetic acid by reacting with the proton from CH₄ dissociation is highly endothermic. To help the process, the labeled O atom needs to be activated from its original position. This activation step overcomes a potential of 1.23 eV. The active oxygen vacancy makes the reaction from CH₄ and CO₂ to acetic acid exothermic. On the other hand, the overall reaction energy from CH₄ and CO₂ to CH₃COOH is still 0.22 eV.

So far we have only considered the possibilities of producing aldehyde and acetic acid following the formation of the acetate species. We understand that other routes may exist which result in dry reforming products or C₂ compounds. For example, CO₂ has been used as a reforming reagent for methane to produce syngas. We calculated the reaction of CO₂ with oxygen vacancy to form adsorbed CO (Zn-CO) or gaseous CO, with O of CO₂ filling the vacancy. The results show that both reactions are endothermic at values of 0.49 eV and 0.76 eV respectively. The generated CO may desorb as a product of reforming reaction or couple with zinc-methyl species to form the acetyl species. The acetyl species could then combine with either the surface hydroxyl to produce acetic acid or surface hydrogen to form acetaldehyde. If methylene could be stabilized on the surface and reacted with the methyl species, an ethyl-derived product such as ethanol could be the likely product. Moreover, aldehyde and ethanol may be produced from hydrogenation and deoxygenation of acetic acid. In the present

study, we focused on the direct coupling reaction of the activated CH₄ with CO₂ and examined the possible products as a consequence of the initial coupling. These results allowed us to develop some preliminary understanding of the complex mechanism of catalytic CO₂ and CH₄ conversion beyond syngas production.

Effect of Metal Dopants on C-C Coupling

We examine other metals, including Al, Cd, In and Ga, as dopant in ceria to catalyze the C-C coupling reaction. These metals are either in the same group with zinc or in the neighboring group in the periodic table. We summarized the results with the structural parameter involved in the reaction in Table 1, and the individual structures are provided in Figure S1.

Table 1. Geometric parameters and calculated results of transition states of C-C coupling over different M (Al, Zn, Cd, In and Ga) - doped ceria catalysts.

Ions	Al	Zn	Cd	In	Ga
C-O distance (Å)	1.19/1.23	1.19/1.24	1.19/1.23	1.19/1.24	1.19/1.24
O-C-O angle(°)	148.9	147.5	148.7	146.9	147.1
M-C distance(Å)	2.43	2.26	2.38	2.49	2.33
C-C distance(Å)	2.32	2.25	2.23	2.17	2.18
Ion radius(Å)	0.54	0.74	0.95	0.80	0.62
electronegativity	1.61	1.65	1.69	1.78	1.81
TS-energy(eV)	0.37	0.51	0.92	1.11	1.21

On different M (Al, Zn, Cd, In and Ga) – doped ceria surface, the CO₂ molecule in all these transition states is activated to almost the same extent. As discussed in the case of Zn-doped ceria, CO₂ is activated from its linear configuration, with two differently elongated C-O bonds. one of

1 the C-O bonds shares the same value of 1.19 Å and the other one is in the range of 1.23 Å~1.24
2 Å. Moreover, the $\angle$ OCO angle of the activated CO₂ is in the range of 145°~150°. The CO₂
3 molecule shared a similar V-shaped structure. However, the M-C and C-C distances in the
4 transition states are different from each other. In addition, the activation barriers are different,
5 with values of 0.37 eV, 0.51 eV, 0.92 eV, 1.11 eV and 1.21 eV for Al, Zn, Cd, Ga and In,
6 respectively.

7 The M-C distance in the transition state ranges from 2.26 Å to 2.49 Å and can be correlated
8 with the ionic radius to some degree. As shown in Table 1, the Al-, Zn- and Cd-doped ceria
9 catalysts follow a general trend that the C-C distance decreases as the barrier increases. In fact,
10 on Ga-doped ceria surface, the activation barrier is 0.1 eV higher than that on In-doped ceria
11 surface. However, the C-C distance on Ga-doped ceria is 2.18 Å, slightly longer than that on
12 In-doped ceria. These results demonstrate that the variation of activation barrier is reflected in
13 the C-C distance in transition state. Further analysis of the results showed that the activation
14 barrier correlates with the electronegativity of the doped metal. The correlation between the C-C
15 coupling barrier and the electronegativity is plotted in Figure 8.

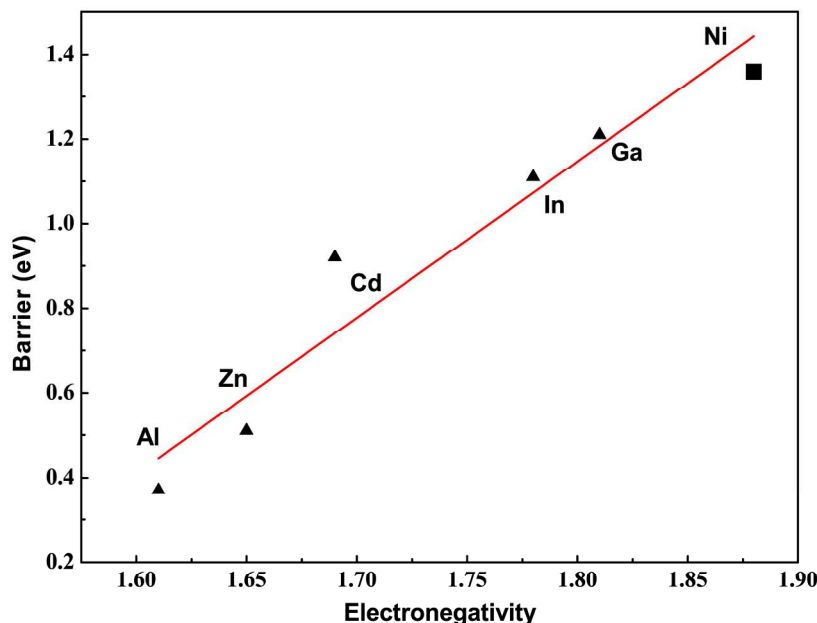


Figure 8. Dependence of the C-C coupling barrier on the electronegativity of the dopants (Al, Zn, Cd, In and Ga).

As shown in Figure 8, the activation barrier for C-C coupling increases as the electronegativity of the doped metal increases. This can be understood from the fact that methyl is stabilized on the surface through the M-C bond. The strength of the M-C bond is expected to be high if the metal has a high electronegativity. The stronger M-C bond will in return reduce nucleophilicity of methyl group, resulting in an increased activation barrier for the coupling reaction. This is reflected in the calculated $\text{CO}_2\text{-CH}_3$ coupling barrier of 1.36 eV by assuming that CH_3 could be stabilized on the Ni-doped ceria. We also plotted the coupling barrier on Ni-doped ceria against the electronegativity of Ni in Figure 8 and found that the linear correlation remained valid. However, doping transition metals would also result in an enhanced dehydrogenation activity. For example, the activation of methane on the Ni-doped ceria to form Ni-CH_3 and coadsorbed H is highly exothermic, with a reaction energy of -1.94 eV. The same reaction on the Ni-doped ceria is endothermic with a reaction of 0.53 eV, higher than the barrier

1 of CO₂ insertion into the CH₃-Zn bond. Furthermore, the dehydrogenation of CH₃ to CH₂ is
2 almost thermally neutral (0.01 eV). These results indicate that CH₃ will not likely to couple with
3 CO₂ before being dehydrogenated to CH₂, and then to CH and C, consistent with the fact that Ni
4 is a good catalyst for dry reforming.⁴⁰ Consequently, we choose base metals instead of transition
5 metals as the active center for the coupling reaction in the present study.

6 7 **CONCLUSIONS**

8 Zn-doped ceria catalyst was found to catalyze the co-conversion of CH₄ and CO₂.
9 Dissociative adsorption of CH₄ is thermodynamically favorable through the formation of a Zn-C
10 bond, with an adsorption energy of -0.53 eV. Our results showed that CO₂ insertion into the Zn-C
11 bond could be achieved under very modest conditions with an activation barrier of 0.51 eV. The
12 insertion is initiated by CO₂ attacking the Zn-CH₃ species from the open side, suggesting a S_E2
13 mechanism. A three-center Zn-C-C bond is formed at the transition state, supported by the DOS
14 analysis. The Brønsted proton generated from CH₄ dissociation could combine with the surface
15 acetate species to form acetic acid although the process is highly endothermic. We also showed
16 that other dopants can also make ceria active for the coupling reaction and the activation barrier
17 correlates with the electronegativity of the doped metal. The present study predicts the possibility
18 of realizing direct C-C coupling between activated CH₄ and CO₂ on heterogeneous catalysts and
19 will have positive impact on the utilization of the greenhouse gases, if implemented in practice.

20 21 **ASSOCIATED CONTENT**

Supporting Information

Relaxed structures involved in the process of product desorption and relaxed structures of C-C coupling on Al-, Cd-, Ga- and In-doped ceria (PDF). This material is available free of charge via the Internet at <http://pubs.acs.org>.

ACKNOWLEDGEMENT

We acknowledge the support of the National Natural Science Foundation of China (Grant Nos. 21373148, 21206116, 21276191, and 21076152). QG acknowledges the support of NSF-CBET program (Award no. CBET-1438440). The high performance computing center of Tianjin University managed and provided major computing resources for this work.

REFERENCES

- (1) (a) Yu, K. M.; Curcic, I.; Gabriel, J.; Tsang, S. C. *ChemSusChem* **2008**, *1*, 893; (b) Aresta, M.; Dibenedetto, A. *Dalton Trans.* **2007**, 2975.
- (2) Metz, B.; Davidson, O.; de Coninck, H.; Loos, M.; Meyer, L.; Eds., *Carbon Dioxide Capture and Storage*. Cambridge University Press: New York, 2005.
- (3) Angamuthu, R.; Byers, P.; Lutz, M.; Spek, A. L.; Bouwman, E. *Science* **2010**, *327*, 313.
- (4) (a) Varghese, O. K.; Paulose, M.; Latempa, T. J.; Grimes, C. A. *Nano Lett.* **2009**, *9*, 731; (b) Sato, S.; Arai, T.; Morikawa, T.; Uemura, K.; Suzuki, T. M.; Tanaka, H.; Kajino, T. *J. Am. Chem. Soc.* **2011**, *133*, 15240.
- (5) (a) Graciani, J.; Mudiyansele, K.; Xu, F.; Baber, A. E.; Evans, J.; Senanayake, S. D.; Stacchiola, D. J.; Liu, P.; Hrbek, J.; Fernandez Sanz, J.; Rodriguez, J. A. *Science* **2014**, *345*, 546; (b) An, X.; Li, J.; Zuo, Y.; Zhang, Q.; Wang, D.; Wang, J. *Catal. Lett.* **2007**, *118*, 264; (c) Huff, C. A.; Sanford, M. S. *J. Am. Chem. Soc.* **2011**, *133*, 18122; (d) Menard, G.; Stephan, D. W. *J. Am. Chem. Soc.* **2010**, *132*, 1796; (e) Courtemanche, M. A.; Legare, M. A.; Maron, L.; Fontaine, F. G. *J. Am. Chem. Soc.* **2014**, *136*, 10708.
- (6) (a) Harshini, D.; Lee, D. H.; Kim, Y.; Nam, S. W.; Han, J. H.; Ham, H. C.; Yoon, C. W. *Catal. Lett.* **2014**, *144*, 656; (b) Faria, E. C.; Neto, R. C. R.; Colman, R. C.; Noronha, F. B. *Catal. Today* **2014**, *228*, 138; (c) Kambolis, A.; Matralis, H.; Trovarelli, A.; Papadopoulou, C. *Appl. Catal., A* **2010**, *377*, 16.
- (7) (a) Pham, M. H.; Goujard, V.; Tatibouët, J. M.; Batiot-Dupeyrat, C. *Catal. Today* **2011**, *171*, 67; (b) Zhang, X.; Dai,

- B.; Zhu, A.; Gong, W.; Liu, C. *Catal. Today* **2002**, *72*, 223.
- (8) (a) Banerjee, A.; Dick, G. R.; Yoshino, T.; Kanan, M. W. *Nature* **2016**, *531*, 215; (b) Beckman, E. J. *Nature* **2016**, *531*, 180.
- (9) (a) Wu, J. F.; Yu, S. M.; Wang, W. D.; Fan, Y. X.; Bai, S.; Zhang, C. W.; Gao, Q.; Huang, J.; Wang, W. *J. Am. Chem. Soc.* **2013**, *135*, 13567; (b) Ding, Y.-H.; Huang, W.; Wang, Y.-G. *Fuel Process. Technol.* **2007**, *88*, 319.
- (10) Nizova, G. V.; Shul'pin, G. B.; Nizova, G. V.; Süß-Fink, G.; Stanislas, S. *Chem. Commun.* **1998**, 1885.
- (11) Kurioka, M.; Nakata, K.; Jintoku, T.; Taniguchi, Y.; Takaki, K.; Fujiwara, Y. *Chem. Lett.* **1995**, 244.
- (12) Huang, W.; Xie, K. C.; Wang, J. P.; Gao, Z. H.; Yin, L. H.; Zhu, Q. M. *J. Catal.* **2001**, *201*, 100.
- (13) Wilcox, E. M.; Roberts, G. W.; Spivey, J. J. *Catal. Today* **2003**, *88*, 83.
- (14) Huang, W.; Zhang, C.; Yin, L.; Xie, K. *J. Nat. Gas Chem.* **2004**, *13*, 113.
- (15) Ding, Y.-H.; Huang, W.; Wang, Y.-G. *Fuel Process. Technol.* **2007**, *88*, 319.
- (16) Liu, C.-j.; Li, Y.; Zhang, Y.-p.; Wang, Y.; Zou, J.; Eliasson, B.; Xue, B. *Chem. Lett.* **2001**, 1304.
- (17) (a) Darensbourg, D. J.; Groetsch, G.; Wiegrefe, P.; Rheingold, A. L. *Inorg. Chem.* **1987**, *26*, 3827; (b) Darensbourg, D. J.; Hanckel, R. K.; Bauch, C. G.; Pala, M.; Simmons, D.; White, J. N. *J. Am. Chem. Soc.* **1985**, *107*, 7463; (c) Johansson, R.; Jarenmark, M.; Wendt, O. F. *Organometallics* **2005**, *24*, 4500; (d) Sakaki, S.; Ohkubo, K. *Inorg. Chem.* **1989**, *28*, 2583; (e) Holl, M. M.; Hillhouse, G. L.; Folting, K.; Huffman, J. C. *Organometallics* **1987**, *6*, 1522; (f) Schmeier, T. J.; Hazari, N.; Incarvito, C. D.; Raskatov, J. A. *Chem. Commun.* **2011**, *47*, 1824.
- (18) (a) Oda, A.; Torigoe, H.; Itadani, A.; Ohkubo, T.; Yumura, T.; Kobayashi, H.; Kuroda, Y. *J. Phys. Chem. C* **2013**, *130917083323008*; (b) Oda, A.; Torigoe, H.; Itadani, A.; Ohkubo, T.; Yumura, T.; Kobayashi, H.; Kuroda, Y. *J. Phys. Chem. C* **2014**, *118*, 15234.
- (19) Panjan, W.; Sirijaraensre, J.; Warakulwit, C.; Pantu, P.; Limtrakul, J. *Phys. Chem. Chem. Phys.* **2012**, *14*, 16588.
- (20) (a) Yeung, C. S.; Dong, V. M. *J. Am. Chem. Soc.* **2008**, *130*, 7826; (b) Ochiai, H.; Jang, M.; Hirano, K.; Yorimitsu, H.; Oshima, K. *Org. Lett.* **2008**, *10*, 2681; (c) Metzger, A.; Bernhardt, S.; Manolikakes, G.; Knochel, P. *Angew. Chem. Int. Ed.* **2010**, *49*, 4665; (d) Kobayashi, K.; Kondo, Y. *Org. Lett.* **2009**, *11*, 2035.
- (21) Sattler, W.; Parkin, G. *J. Am. Chem. Soc.* **2011**, *133*, 9708.
- (22) Cheng, Z.; Sherman, B. J.; Lo, C. S. *J. Chem. Phys.* **2013**, *138*, 014702.
- (23) (a) Kresse, G.; Hafner, J. *Phys. Rev. B* **1993**, *48*, 13115; (b) Kresse, G.; Furthmüller, J. *Phys. Rev. B* **1996**, *54*, 11169.
- (24) Perdew, J. P.; Burke, K.; Ernzerhof, M. *Phys. Rev. Lett.* **1996**, *77*, 3865.
- (25) Cheng, L.; Mei, D.; Ge, Q. *J. Phys. Chem. C* **2009**, *113*, 18296.
- (26) (a) Mayernick, A. D.; Janik, M. J. *J. Catal.* **2011**, *278*, 16; (b) Mayernick, A. D.; Janik, M. J. *J. Phys. Chem. C* **2008**, *112*, 14955; (c) Krcha, M. D.; Mayernick, A. D.; Janik, M. J. *J. Catal.* **2012**, *293*, 103; (d) Dholabhai, P. P.; Adams, J. B.; Crozier, P.; Sharma, R. *J. Chem. Phys.* **2010**, *132*, 094104; (e) Vecchietti, J.; Bonivardi, A.; Xu, W.; Stacchiola, D.; Delgado, J. J.; Calatayud, M.; Collins, S. E. *ACS Catal.* **2014**, *4*, 2088.
- (27) (a) Kresse, G.; Joubert, D. *Phys. Rev. B* **1999**, *59*, 1758; (b) Blöchl, P. E. *Phys. Rev. B* **1994**, *50*, 17953.
- (28) (a) Kümmerle, E. A.; Heger, G. *J. Solid State Chem.* **1999**, *147*, 485; (b) Vilé, G.; Dähler, P.; Vecchietti, J.; Baltanás, M.; Collins, S.; Calatayud, M.; Bonivardi, A.; Pérez-Ramírez, J. *J. Catal.* **2015**, *324*, 69.
- (29) Hsu, L.-C.; Tsai, M.-K.; Lu, Y.-H.; Chen, H.-T. *J. Phys. Chem. C* **2013**, *117*, 433.
- (30) Knapp, D.; Ziegler, T. *J. Phys. Chem. C* **2008**, *112*, 17311.
- (31) Ye, J.; Liu, C.; Mei, D.; Ge, Q. *ACS Catal.* **2013**, *3*, 1296.
- (32) Patil, U.; Saih, Y.; Abou-Hamad, E.; Hamieh, A.; Pelletier, J. D.; Basset, J. M. *Chem. Commun.* **2014**, *50*, 12348.

- (33) Fan, T.; Chen, X.; Lin, Z. *Chem. Commun.* **2012**, *48*, 10808.
- (34) Johnson, M. T.; Johansson, R.; Kondrashov, M. V.; Steyl, G.; Ahlquist, M. S.; Roodt, A.; Wendt, O. F. *Organometallics* **2010**, *29*, 3521.
- (35) Mankad, N. P.; Gray, T. G.; Laitar, D. S.; Sadighi, J. P. *Organometallics* **2004**, *23*, 1191.
- (36) Yang, X.-F.; Wang, A.; Qiao, B.; Li, J.; Liu, J.; Zhang, T. *Acc. Chem. Res.* **2013**, *46*, 1740.
- (37) Ye, J.; Liu, C.; Ge, Q. *Phys. Chem. Chem. Phys.* **2012**, *14*, 16660.
- (38) Zhang, R.; Song, L.; Liu, H.; Wang, B. *Appl. Catal., A* **2012**, *443-444*, 50.
- (39) Calaza, F. C.; Chen, T.-L.; Mullins, D. R.; Xu, Y.; Overbury, S. H. *Catal. Today* **2015**, *253*, 65.
- (40) Liu, Z.; Grinter, D. C.; Lustemberg, P. G.; Nguyen-Phan, T. D.; Zhou, Y.; Luo, S.; Waluyo, I.; Crumlin, E. J.; Stacchiola, D. J.; Zhou, J.; Carrasco, J.; Busnengo, H. F.; Ganduglia-Pirovano, M. V.; Senanayake, S. D.; Rodriguez, J. A. *Angew. Chem. Int. Ed.* **2016**, *55*, 7455.

Table of Contents

